Agnes Teresa Davuri

AI / ML Research Engineer · Causal ML · Agentic LLM Systems · Reinforcement Learning · Quantum

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Computer Science graduate (NIT Warangal, 2026) who builds **complete, empirically validated AI systems**: causal ML, agentic LLM pipelines, reinforcement learning and quantum computing. **Ten independent research projects**, each with public code, real data and honestly reported results, including the negative ones.

10	589K	280+	AIR 1364
independent research projects, all with public code	real Lending Club loans analysed with causal methods	circuit runs on real IBM 156-qubit hardware	JEE Mains 2022 of 1.2 million candidates

RESEARCH PROJECTS

Causal-RAG — Counterfactual-Audited Retrieval for High-Stakes Lending — 2026

Four-agent system built on CALF, run on 589,099 real Lending Club loans plus Fannie Mae / CFPB policy text

- Designed four cooperating agents (retriever, causal attribution, counterfactual perturbation, audit) that return APPROVE / DECLINE / REFER decisions with a causal DAG and JSON audit trail; realised default rates order cleanly at 11.5% / 17.7% / ~23%.
- Validated causal effects without ground truth: calibration slope 0.86–1.18 on 3 of 4 underwriting clauses (LightGBM T-learner: 0.04–0.27), ATE inside a real randomised experiment's interval, all refutation tests passed.
- Control experiments showed causal evidence added no measurable AUROC over a flags-only baseline (0.678 vs 0.668), and that cutoff-style prompts make LLM decisions step-like (cliff index 86 vs 1.08). Reported openly.

Qwen2.5-7B (4-bit) · PyTorch · LightGBM · causal-learn · Gradio | Code: github.com/Davuri124/Casual-RAG

Budget-Constrained RL for Human-in-the-Loop Agent Orchestration — 2026

Real-data extension of RLHIL: a policy that decides when to automate, re-retrieve, call a stronger model or ask a human, under a hard review budget

- Built a real-execution replay environment on HotpotQA (no synthetic data) where a PPO orchestrator works under a hard cap of K human reviews per 1,000 queries: 0 budget violations, full SQLite audit trail.
- Adding a supervised risk score to the policy state beat the plain RL policy in all 15 stream × budget settings (+0.005 to +0.021 net reward) and the strongest paced baseline in 4 (10 ties, 1 loss); designed a pre-registered, Holm-corrected confirmatory test on 2,000 fresh queries.

PPO · vLLM · Qwen2.5-3B/7B · DeBERTa NLI · gradient-boosted risk models | Code: github.com/Davuri124/hitl-orchestrator

SOLAR — Label-Efficient Evaluation of Detectors and LLM Judges — 2026

Started as a seven-agent misinformation ensemble; audited it, then rebuilt the project around whether any detector can be trusted

- Showed detectors do not transfer: cross-dataset AUC fell 0.11–0.27 on 6 of 7 bot datasets, and no bot-share estimator beat guessing the average.
- Built a small-sample toolkit: ~200 human labels give 95% intervals (0.93–0.96 coverage) with ~10x lower error than counting detector flags (2.7 vs 28.9 points); measured that GPT-4 disagrees with the human majority on 28.7% of 1,612 MT-Bench items.
- Built an always-valid monitor that missed the truth in 2–3% of streams vs 42–46% for ordinary intervals; shipped two browser tools. Fixed success bars before every run and reported the failures.

Python · Qwen2.5-3B · prediction-powered inference · confidence sequences · FastAPI | Code: github.com/Davuri124/SOLAR

CALF — Causally-Aware Layered Fabric for Treatment Effect Estimation — 2026

Tree-structured causal neural architecture, applied to insurance underwriting

- Encoded covariates into four causally ordered layers fused by cross-layer attention and proved nonparametric identification.
- Lowest PEHE among five baselines on an NLSY97 benchmark ($p < 0.001$ vs Causal Forest); reported the null result on insurance underwriting plainly.

PyTorch · causal inference | Code: github.com/Davuri124/CALF

RLHIL — RL for Human-in-the-Loop Intervention — 2025 – 2026

Budget-constrained CMDP for adaptive AI oversight

- Formalised intervention timing as a Constrained MDP (8-dim state, 4-action multi-expert selection, PPO): ~96% decision boundary accuracy, regret well below the theoretical bound.
- Curriculum learning gave a large gain on hard pipelines; validated on 4 real-world datasets (fraud, medical, financial) with bootstrap CIs; outperforms Learning-to-Defer and confidence-threshold deferral.

PPO · constrained MDPs · Stable-Baselines3 | Code: github.com/Davuri124/Human_Intervention_Points

QuLit — Quantum Encoding of Multilingual Literary Text — 2025 – 2026

Density-matrix semantics for semantic ambiguity across 6 languages, validated on real IBM Quantum hardware

- Encoded multilingual literary text as quantum density matrices via 4-qubit parameterised circuits, beating mBERT on ambiguity resolution and cross-lingual similarity.

PennyLane · Qiskit · IBM Quantum

HERMES — Reader-Adaptive Multilingual Literary Interpretation — 2025 – 2026

Extends QuLit with a reader-conditioned model (culture × era × emotion)

- Strong cross-cultural alignment ($r = 0.79$, beating mBERT); reports its weaker results (decoder collapse, one metric not yet significant) plainly.

PyTorch · HuggingFace Transformers | Code: github.com/Davuri124/HERMES

Grover Noise Fingerprinting (GNF) — Noise-Optimal Grover on NISQ Devices

2025 – 2026

Formal theorem and hardware readiness metric, IBM Quantum

- Derived the noise-optimal Grover iteration count k*(N,ε) as a formal theorem with a closed form (2.4% error vs numerics).
- Introduced zero-overhead noise extraction using Grover as its own probe plus the Grover Fidelity Score; across 280+ executions on two IBM 156-qubit backends it gave a real +28.4% success-probability gain.

Qiskit · IBM Quantum | Code: github.com/Davuri124/Grover-Algorithm-on-NISQ-devices

Osmotic Attention — Information-Gradient Context Flow in Transformers

2026

Bio-inspired attention mechanism, 11 experimental phases

- Added an information-density-gradient term to attention logits; tested on GLUE, SQuAD, long-range and cross-lingual/cross-modal tasks with 5 seeds and bootstrap CIs.
- Honest mixed result: consistent positive signal for from-scratch cross-attention on QA, a small real negative effect when injected into pretrained BERT.

PyTorch · HuggingFace | Code: github.com/Davuri124/osmotic-attention

Predictive Quantum Governance System

2025 – 2026

Metadata-driven AI orchestration with quantum-assisted decisions

- Built a governance pipeline validated on 5 real datasets (Amazon Reviews, Online Retail, MIMIC ICU, IEEE Fraud, Amazon Multi-Vendor); Governance Confidence Index gives 27–28% risk reduction across domains.
- Quantum-assisted decisions via a 3-tier Qiskit fallback (IBM Quantum → Aer → StatevectorSampler); real-time Kafka streaming governance.

Qiskit · Apache Kafka · Python | Code: github.com/Davuri124/Metadata_Orchestration_Pipeline

EXPERIENCE

Intarinz — Data Engineering Intern, AWS data pipelines

May – Jul 2024

- Built end-to-end AWS data pipelines (S3, Glue, Lambda, Redshift) for structured and semi-structured enterprise data ingestion.
- Automated transformation, quality validation and lineage tracking; improved monitoring and alerting to cut manual intervention.

TECHNICAL SKILLS

Languages	Python, C++, Java, SQL, HTML, CSS, LaTeX
ML / AI	PyTorch, TensorFlow, Scikit-learn, LightGBM, HuggingFace Transformers, sentence-transformers, Stable-Baselines3, PennyLane, Qiskit
LLMs & agents	Qwen2.5 (3B/7B), vLLM, 4-bit quantisation (bitsandbytes, AWQ), RAG, multi-agent pipelines, LLM evaluation and auditing, Gradio
Reinforcement learning	PPO, PPO-Lagrangian, constrained MDPs, action masking, cached-counterfactual replay environments
Causal inference & stats	CALF, EconML, doubly-robust (AIPW) calibration, propensity matching, PC causal discovery, refutation tests, E-values, paired bootstrap, pre-registered evaluation, prediction-powered inference, confidence sequences
Cloud / data	AWS (S3, Glue, Lambda, Redshift), Apache Kafka, IBM Quantum Platform, PostgreSQL, MongoDB, Neo4j, SQLite
Tools	Git, GitHub, VS Code, Jupyter, Google Colab, Kaggle, Weights & Biases, Streamlit, FastAPI

EDUCATION & ACHIEVEMENTS

National Institute of Technology, Warangal — B.Tech, Computer Science and Engineering

2022 – 2026

Jawahar Navodaya Vidyalaya — Class XII (2021–2022), Class X (2019–2020), CBSE

- JEE Mains 2022: All India Rank 1364 (CRL), Category Rank 26, out of 1.2 million candidates
- IBM Certifications: Web Development; Working in a Digital World: Professional Skills